



KZY-1000

Zirconia Y-TZP

KEY PROPERTIES

- High strength
- High toughness
- High wear resistance under sliding conditions
- Low friction coefficient against metals
- Excellent surface finish ($R_a < 0.05$)

APPLICATIONS

- Structural parts for machinery
- Welding pins
- Plungers for high pressure pumps
- Metal packaging tooling
- Solids handling systems
- High pressure nozzles

MAIN PROPERTIES

Property	Value	Unit
Density	6.0	g/cm ³
Flexural strength	1000	MPa
Fracture Toughness	6.0	MPa·√m
Hardness	13	GPa
Young Modulus	210	GPa
Thermal Conductivity	3	W/m·K
CTE (20–1000°C)	11	x10 ⁻⁶ K ⁻¹
Electrical Resistivity	>10 ¹¹	Ω·cm

* All properties measured at 20°C unless otherwise stated.

** Measured in a wall thickness of 2 mm

OPERATING CONDITIONS

Max working temp (air)		900	°C
Recomended temp		<850	°C
Continuous temp		800	°C

ELECTRICAL BEHAVIOR

Insulator			
Electrical resistivity		>10 ¹¹	Ω·cm

MATERIAL INFORMATION

Material type		Y-TZP zirconia
Structure		Fully stabilized ceramic
Key feature		High strength + wear resistance
Machinability		Medium
Color		White

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

The information provided is for guidance only and does not constitute a guarantee.

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